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MAXIMAL LIFT-DRAG RATIO OF WING-CONE AND WING-HALF-CONE COMBINATIONS AT HIGH SUPERSONIC VELOCITIES

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The use of similarity relations in an inviscid gas at Mach number $M_\infty \geq 4$ of the oncoming flow makes it possible to represent the maximal lift-drag ratio $K_{\max}$ and the corresponding lift coefficient $C_{y \text{ opt}}$ of combinations of a flat infinitesimally thin delta wing and a cone (symmetric configuration) or half-cone on the upper or lower surface of the wing (asymmetric configurations) as functions of two variables. The regions of relative advantage with respect to $K_{\max}$ are constructed by numerical solution of the Euler equations for the investigated configurations, and the influence of the viscosity of the gas on the shape of these regions is considered.

1. Experimental and theoretical investigations of the maximal lift-drag ratio of conical wing-fuselage combinations, which are idealizations of possible hypersonic aircraft, show that at high supersonic velocities the largest value of $K_{\max}$ is attained in different regimes for different relative position of the wing and fuselage [1-3].

It follows from the similarity relations of [5] that in an inviscid gas when $M_\infty \geq 4$, sufficiently slender conical configurations ($n = (M_\infty^2 - 1)^{1/2} \tan \delta \leq 1$, where δ is the half-angle of the cone), and for sufficiently small angles of attack that nevertheless include the $K_{\max}$ regime the following relations are satisfied to an accuracy of not less than 3%:

$$M_\infty / K_{\max} = \Psi(\mu, m), \quad M_\infty (M_\infty^2 - 1)^{1/2} C_{y \text{ opt}} = I_{\text{opt}}(\mu, m), \quad \mu = (M_\infty^2 - 1)^{1/2} V^{2/3} / S, \quad m = (M_\infty^2 - 1)^{1/2} \tan \beta$$

Here, V is the volume of the configuration, S is the area of the wing, and β is the angle at the tip of the wing.

The reduced volume coefficient μ [5] is a similarity parameter and at the same time preserves the properties of the volume coefficient $\tau = V^{2/3} / S$, which is widely used to compare the aerodynamics of different schemes of hypersonic aircraft. For the investigated configurations, there exists a simple connection between the three parameters μ , m , and n :

$$\mu = \left(\frac{\pi}{3 \cdot 2^j} \right)^{1/2} \frac{n^{4/3}}{m} \quad (1.1)$$

where $j = 0$ for a wing with cone, and $j = 1$ for a wing with half-cone.

The numerical calculations of [4] have been used to plot on the μ, m plane the regions of favorable interference (Figs. 1-4), which show which of the considered configurations in the various ranges of variation of the determining parameters have the largest maximal lift-drag ratio (at equal volumes). The boundaries separating these regions are determined with accuracy $\sim 10\%$, and are therefore somewhat smeared. The condition of equality of the angles at the tip of the wing and the half-angle of the

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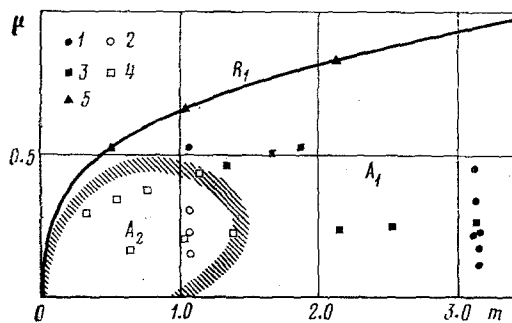


Fig. 1

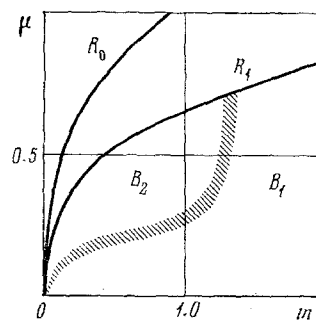


Fig. 2

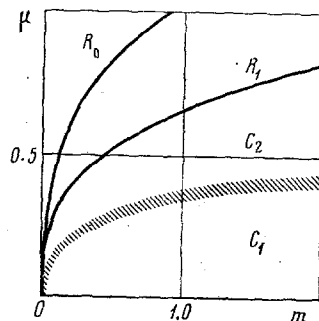


Fig. 3

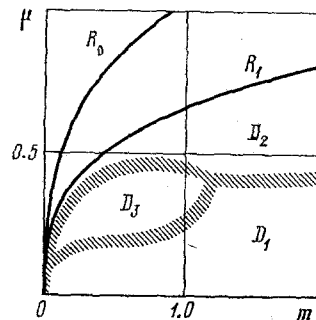


Fig. 4

cone or half-cone ($m = n$) determines the boundaries of the region of existence of the solutions on the μ, m plane (see (1.1)). These curves, which are denoted by R_0 and R_1 for the symmetric and asymmetric configurations, respectively, are also plotted in Figs. 1-4. The regions of parameter values for which solutions exist are situated below these curves. The axis $\mu = 0$ corresponds to an isolated wing, and the point $m = 1$ on it to sonic leading edges of the wing.

In Fig. 1, we consider a wing with half-cone on its upper and lower surfaces. The maximal lift-drag ratio of the former exceeds $K_{\max}$ of the latter in the region A_1 . In the region A_2 , the situation is reversed. The maximal lift-drag ratio of a half-cone with flat base is always greater than $K_{\max}$ of the inverted configuration (curve R_1). The points in Fig. 1 correspond to the results of experiments that will be discussed below.

A wing with cone and a wing with half-cone on its lower surface are compared in Fig. 2. Between the curves R_0 and R_1 there is a region in which solutions exist only for the symmetric configuration; below the curve R_1 , solutions are possible for both the considered combinations. In the region B_1 , the value of $K_{\max}$ for the symmetric configuration is greater than for the asymmetric; in region B_2 , the opposite is true. For $m < 0.4$, the position of the boundary agrees with the results obtained from slender-body theory [3]. The maximal lift-drag ratio of a wing with cone exceeds the $K_{\max}$ of a wing with half-cone on the upper surface in the region C_1 and is less than it in the region C_2 (Fig. 3).

The superposition of the regions discussed above is shown in Fig. 4. In the region D_1 , the wing-cone configuration has the greatest maximal lift-drag ratio among all the investigated configurations; in the region D_2 , it is the wing with a half-cone on its upper surface, and in D_3 one with the half-cone on its lower surface. At the point $\mu \approx 0.4$, $m \approx 1.2$ the maximal lift-drag ratios of the considered configurations are equal.

Analysis of the numerical calculations shows that the lift l_{opt} of the two asymmetric configurations is the same and somewhat greater than for the symmetric configuration.

2. To use the obtained results to investigate the aerodynamic characteristics of the considered combinations with allowance for viscosity, we exploit the fact that,

on the basis of the results of the linear theory [6], the wave drag polar can be represented as a quadratic trinomial whose coefficients can be expressed in terms of the $K_{\max}$ of the combination in the normal and inverted position and $C_{y \text{ opt}}$. The numerical calculations even for bodies with strong asymmetry and appreciable volume confirm this well at the considered Mach numbers M_∞ and C_y values not appreciably exceeding $C_{y \text{ opt}}$.

Comparison of the calculated and experimental dependences of the coefficient of the axial force obtained for the investigated configurations shows that at Reynolds number $Re \geq 10^6$ the increase in the drag coefficient C_x due to the viscosity of the gas is almost independent of the angle of attack in the range of variation of this quantity in which we are interested.

From these approximate but well-satisfied relations we can draw a conclusion about the influence of viscosity on the shape of the regions of favorable interference. Since the corresponding values of the lift $C_{y \text{ opt}}$ for the two asymmetric configurations are the same and the increments in C_x due to viscosity for them are equal, the boundary between the regions A_1 and A_2 of favorable interference keeps its position in a viscous gas as well. In Fig. 1, we have plotted the experimental points 1 and 2 from [1] and 5 from [2], the experimental data obtained by V. V. Keldysh and the author ([3, 4]) for sweepback angles of the leading edge of the wing $\chi = 79^\circ$ and 70° , $\delta = 8^\circ$, $M_\infty = 2, 3, 4, 6, 7, 8.7, 9.7$. Points 1, 3, and 5 correspond to larger values of $K_{\max}$ of the wing with half-cone on the upper surface, and points 2 and 4 to values that are smaller compared with the inverted configuration. The agreement between the calculated and experimental results in the range $M_\infty = 2-20$ is good.

Because of the smaller values of $C_{y \text{ opt}}$ and the usually larger coefficients of friction for the symmetric configurations compared with the asymmetric, the regions B_2 and C_2 in Figs. 2 and 3 are somewhat expanded at the expense of regions B_1 and C_1 by the allowance for the viscosity of the gas, the change in the position of the boundaries depending on the state of the boundary layer and the Reynolds number. Nevertheless, Figs. 2-4 give a qualitative picture of the regions of favorable interference for the considered configurations in a viscous gas as well.

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